

31.0.1 NIELIT 2017 July Scientist B (CS) - Section B: 24



How many wires are threaded through the cores in a coincident-current core memory?

- A. 2 B. 3 C. 4 D. 6

nielit2017july-scientistb-cs non-gatecse

Answer key

31.0.2 NIELIT 2016 MAR Scientist B - Section C: 33



In what module multiple instances of execution will yield the same result even if one instance has not terminated before the next one has begun?

- A. Non reusable module B. Serially usable
C. Re-enterable module D. Recursive module

nielit2016mar-scientistb non-gatecse

31.0.3 NIELIT 2016 MAR Scientist C - Section B: 18



Find the volume of the solid obtained by rotating the region bound by the curves $y = x^3 + 1$, $x = 1$, and $y = 0$ about the x -axis

- A. $\frac{23\pi}{7}$ B. $\frac{16\pi}{7}$ C. 2π D. $\frac{19\pi}{7}$

nielit2016mar-scientistc non-gatecse

31.0.4 NIELIT 2016 MAR Scientist C - Section B: 15



The equation of the plane through the point $(-1, 3, 2)$ and perpendicular to each of the planes $x + 2y + 3z = 5$ and $3x + 3y + z = 0$ is

- A. $7x - 8y + 3z + 25 = 0$ B. $7x + 8y + 3z + 25 = 0$
C. $7x - 8y + 3z - 25 = 0$ D. $7x - 8y - 3z - 25 = 0$

nielit2016mar-scientistc non-gatecse

31.0.5 NIELIT 2016 MAR Scientist C - Section B: 14



Find the area bounded by the curve $y = \sqrt{5 - x^2}$ and $y = |x - 1|$

- A. $\frac{2}{0}(2\sqrt{6} - \sqrt{3}) - \frac{5}{2}$ B. $\frac{2}{3}(6\sqrt{6} + 3\sqrt{3}) + \frac{5}{2}$
C. $2(\sqrt{6} - \sqrt{3}) - 5$ D. $\frac{2}{3}(\sqrt{6} - \sqrt{3}) + 5$

nielit2016mar-scientistc non-gatecse

31.0.6 NIELIT 2016 MAR Scientist C - Section B: 7



The area under the curve $y(x) = 3e^{-5x}$ from $x = 0$ to $x = \infty$ is

- A. $\frac{3}{5}$ B. $\frac{-3}{5}$ C. 5 D. $\frac{5}{3}$

nielit2016mar-scientistc non-gatecse

31.0.7 NIELIT 2016 MAR Scientist C - Section B: 6



If $y = f(x)$, in the interval $[a, b]$ is rotated about the x -axis, the Volume of the solid of revolution is $(f'(x) = dy/dx)$

- A. $\int_a^b \pi [f(x)]^2 dx$ B. $\int_a^b [f(x)]^3 dx$
C. $\int_a^b \pi [f'(x)]^2 dx$ D. $\int_a^b \pi^2 f(x) dx$

nielit2016mar-scientistc non-gatecse

31.1 Complex Number (1)

31.1.1 Complex Number: NIELIT 2018-28



For the function $(z) = \frac{1}{z^2(e^z - 1)}$, $z = 0$ is a pole of order:

- A. 1 B. 2 C. 3 D. None of these

nielit-2018 non-gatecse complex-number

Answer key

31.2 Differential Equation (4)

31.2.1 Differential Equation: NIELIT 2018-18



If y^a is an integrating factor of the differential equation $2xydx - (3x^2 - y^2)dy = 0$, then the value of a is

- A. -4 B. 4 C. -1 D. 1

nielit-2018 non-gatecse differential-equation

Answer key

31.2.2 Differential Equation: NIELIT 2018-20



The general solution of the differential equation $\frac{dy}{dx} = (1 + y^2)(e^{-x^2} - 2x \tan^{-1} y)$ is:

- A. $e^{x^2} \tan^{-1} y = x + c$ B. $e^{-x^2} \tan y = x + c$

C. $e^x \tan y = x^2 + c$

D. $e^{-x} \tan^{-1} y = x^3 + c$

nielit-2018 non-gatecse differential-equation

31.2.3 Differential Equation: NIELIT 2018-22



While solving the differential equation $\frac{d^2y}{dx^2} + 4y = \tan 2x$ by the method of variation of parameters, then value of Wronskian (W) is:

A. 1

B. 2

C. 3

D. 4

nielit-2018 non-gatecse differential-equation

Answer key

31.2.4 Differential Equation: NIELIT 2018-25



The general solution of the partial differential equation $(D^2 - D'^2 - 2D + 2D')Z = 0$ where $D = \frac{\partial}{\partial x}$ and $D' = \frac{\partial}{\partial y}$:

A. $f(y+x) + e^{2x}g(y-x)$

B. $e^{2x}f(y+x) + g(y-x)$

C. $e^{-2x}f(y+x) + g(y-x)$

D. $f(y+x) + e^{-2x}g(y-x)$

nielit-2018 non-gatecse differential-equation partial-order

31.3 Double Integration (1)

31.3.1 Double Integration: NIELIT 2018-15



The value of $\int_1^2 \int_0^{x^2} x \, dy \, dx$ is:

A. $\frac{15}{4}$

B. $\frac{5}{4}$

C. $\frac{3}{4}$

D. $\frac{2}{3}$

nielit-2018 calculus double-integration

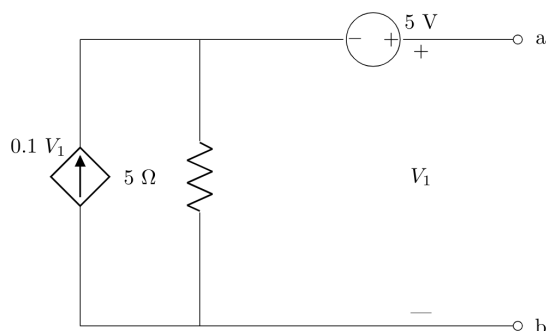
Answer key

31.4 Electronic Circuit (1)

31.4.1 Electronic Circuit: NIELIT Scientist B 2020 November: 71



The resistance to be connected across terminal a, b for maximum power transfer to it is:



A. $40\ \Omega$ B. $5\ \Omega$ C. $2.5\ \Omega$ D. $10\ \Omega$

nielit-scb-2020 electronic-circuit non-gatecse

31.5

General (1)

31.5.1 General: NIELIT 2023 Feb Scientist C | Section C | Question: 75



Which of the following is a characteristic of qualitative research?

A. Design flexibility

B. Inductive analysis

C. Context sensitivity

D. All of the above

nielit2023feb-scientistc general

Answer key

31.6

Integration (2)

31.6.1 Integration: NIELIT 2018-16



The value of $\int_c \frac{2x^2-5}{(x+2)^2(x^2+4)x^2} dx$, (where c is the square with vertices $1+i, 2+i, 2+2i, i+2i$) is:

A. 0

B. πi C. $2\pi i$ D. $4\pi i$

nielit-2018 non-gatecse integration

31.6.2 Integration: NIELIT 2018-29



Using Green's theorem in plane, evaluate $\int_c (2x-y)dx + (x+y)dy$, where c is the circle $x^2 + y^2 = 4$ in the plane:

A. 2π B. 4π C. -4π D. 8π

nielit-2018 non-gatecse integration

Answer key

31.7

Partial Order (3)

31.7.1 Partial Order: NIELIT 2016 MAR Scientist C - Section B: 5



If $f(x, y) = x^3y + e^x$, the partial derivatives, $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ are

A. $3x^2y + 1, x^3 + 1$ B. $3x^2y + e^x, x^3$ C. $x^3y + xe^x, x^3 + e^x$ D. $2x^2y + \frac{e^x}{x}$

nielit2016mar-scientistc non-gatecse partial-order

Answer key

31.7.2 Partial Order: NIELIT 2018-19



If $u = f(y - z, z - x, x - y)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ is equal to:

- A. $x + y + z$ B. $1 + x + y + z$ C. 1 D. 0

nielit-2018 non-gatecse differentiation partial-order

31.7.3 Partial Order: NIELIT 2018-23



If $w = f(z) = u(x, y) + i v(x, y)$ is an analytic function, then $\frac{dw}{dz}$ is:

- A. $\frac{\partial u}{\partial x} - i \frac{\partial u}{\partial y}$
 B. $\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial y}$
 C. $\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x}$
 D. $\frac{\partial u}{\partial x} + i \frac{\partial u}{\partial y}$

nielit-2018 non-gatecse differentiation partial-order

31.8 Permutation and Combination (1)

31.8.1 Permutation and Combination: NIELIT 2021 Dec Scientist B - Section A: 23



The students in a class are seated according to their marks in the previous examination. Once it so happens that four of these students get equal marks and therefore the same rank. To decide their seating arrangement, the teacher wants to write down all possible arrangements, one in each of separate bits of paper, in order to choose one of these by lots. How many bits of paper are required ?

- A. 9 B. 12 C. 15 D. 24

nielit2021dec-scientistb counting permutation-and-combination analytical-aptitude

Answer key 

Answer Keys

31.0.1	A	31.0.2	C	31.0.3	TBA	31.0.4	A	31.0.5	TBA
31.0.6	A	31.0.7	A	31.1.1	B	31.2.1	A	31.2.2	C
31.2.3	B	31.2.4	A	31.3.1	A	31.4.1	B	31.5.1	D
31.6.1	A	31.6.2	A	31.7.1	B	31.7.2	C	31.7.3	A
31.8.1	D								